POLITECNICO
MILANO 1863

Tutoring Section 2 Class 1
Numerical Problems in Machining

Dr. Huan Shao

Manufacturing and Production Systems Research Line
Mechanical Engineering Department, Politecnico di Milano

17:30 – 19:00 , 25/Nov/2022

huan.shao@polimi.it

- **Problem 1 (15/Jan/2021 exam)**

A steel bar with a diameter of 30 mm and a length of 160 mm is hold by means of a self-centering three-jaw chuck and tailstock (dead centre). The workpiece undergoes external cylindrical turning, which brings the diameter to the final value of 24 mm over a length of 90 mm. The machining is performed in two steps: a roughing step and a finishing step. Considering the roughing step, whose depth of cut is equal to 2 mm, determine:

Q1. The values of cutting force and cutting power for the roughing step, considering a value of $k_{c0,4}$ equal to 2150 MPa, a coefficient x of the Kronenberg relationship equal to 0,29 and a value of main entering angle equal to 90° . The recommended feed is 0,15 mm/rev, while the cutting speed is 260 m/min.

Q2. The optimal tool life and the corresponding cutting speed, in case you want to optimize the production time. The tool change time T_t is equal to 120 seconds, while the constants of the Taylor's law are C = 250 and n = 0,122.

- **Problem 2 (10/Sep/2021 exam)**

A steel bar with a diameter of 100 mm and a length of 300 mm is hold by means of a self-centering three-jaw chuck and tailstock (dead centre). The workpiece undergoes external cylindrical turning, which brings the diameter to the final value of 94 mm over a length of 180 mm. The machining is performed in two steps: a roughing step and a finishing step. Considering the roughing step, whose depth of cut is equal to 2,5 mm, determine:

Q3. The values of cutting force and cutting power, taking into account a value of $k_{c0,4}$ equal to 2100 MPa, a coefficient x of the Kronenberg relationship equal to 0,29 and a value of main entering angle equal to 45° . The recommended feed is 0,15 mm/rev while the cutting speed is 260 m/min.

Q4. The optimal tool life and the corresponding cutting speed in case you want to optimize the production cost. The tool changing time is equal to 160 seconds, the cutting-edge cost is equal to 1 €, and the operating cost of the machine tool is 180 €/h, while the constants of the Taylor relation are C = 250 and n = 0,122.